Comparative Analysis of Language Production in Late-vs-Typical Talking Toddlers

💡 PS: please read this document ***before*** going over the `jupyter` notebook. it contains a comprehensive background of the project as well as details of each aspect (it reuses few parts of the powerpoint slides for this); whereas the notebook contains the code, explanatory comments wherever required, and the output tables and figures

Introduction and Data Selection

The project stems from an interest in understanding child language acquisition, in particular the developmental progression through which a child learns language, a progression which seems to be guided by a species-specific biology (Gleitman and Newport, 1995). This progression would be best captured by longitudinal studies that contain linguistic data of conversations with toddlers once they begin speaking two-three word utterances, up to when they develop as young children and are able to communicate longer and more complex sentences.

The CHILDES language database is a rich archive of corpora of speech data of many children with their interlocutors. After an analysis of the various corpora available, I chose the Eliss Weismer Corpus as the dataset for the project because apart from fulfilling the longitudinal linguistic study criteria, it also checked other important boxes for a robust and good dataset. For instance, it has a good sample size (`N = 138`), consistent data collection practices (engage child but limit adult talk, avoid yes/no questions, etc.), and a clear documentation of the nomenclature used in the data files.

The other datasets which were explored are outlined below for reference:

- https://childes.talkbank.org/access/Clinical-Eng/ENNI.html
    - age 4-9
    - N = 77 LI (language impaired) and 300 control

    - amount of data is big yet does lacks similar representation of the two groups
- Language Production in 24-Month-Old Inner-City Children of Cocaine-and-OtherDrug-Using Mothers
    - N is too small and methodology is culture and context specific
- https://childes.talkbank.org/access/Eng-NA/Bates.html
    - N = 27 (too small) and age range is narrow (children at 10, 13, 20, and 28 months)
- https://childes.talkbank.org/access/Eng-NA/Champaign.html
    - N = 44 and age varies from 21, 24, 27, 30, 33, and 36 months

Research Question

The guiding research question is to understand what are the differences in language production between late-talking-toddlers (due to language impairment) vs toddlers who follow a typical developmental journey? Are the differences noticeable/significant? Analysing this difference seeks to shed light on the developmental progression of typical language acquisition.

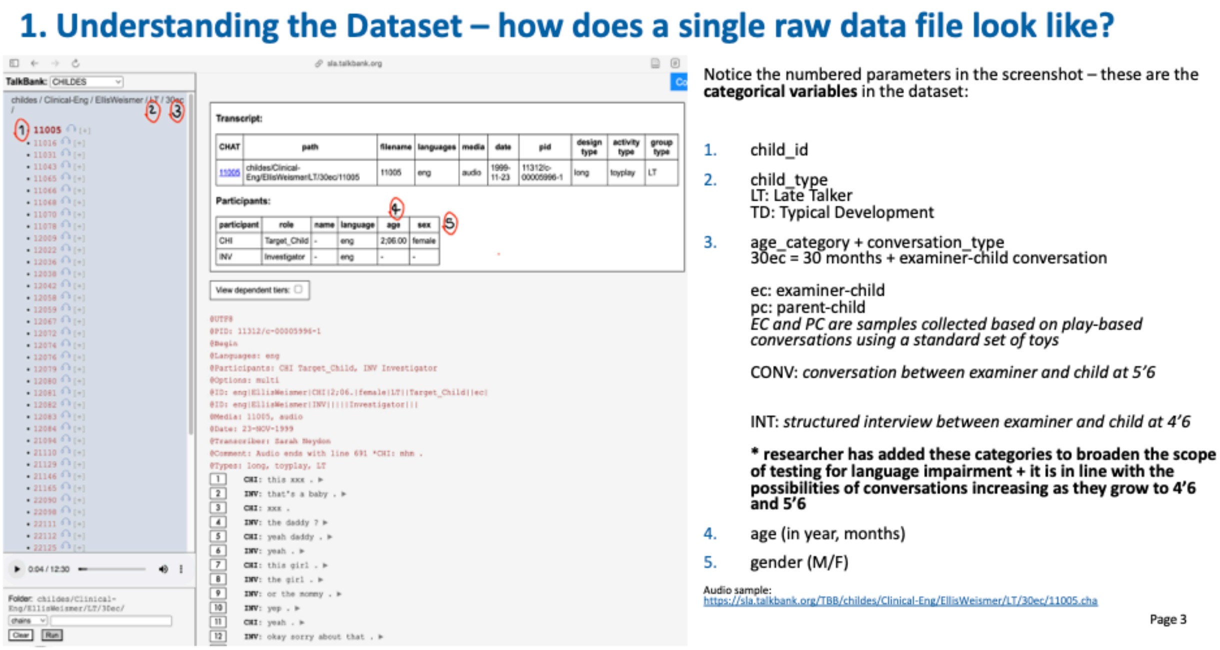
The nature of language data processing and the developmental measures that are important in assessing language production were novel for me in this project. Hence, a hypothesis could not be laid out at the start. The data analysis was exploratory in trying to extract the relevant information (even figuring out *what* is relevant) from the given linguistic corpora, and in assessing the difference in language production between the two groups.

Understanding the Dataset

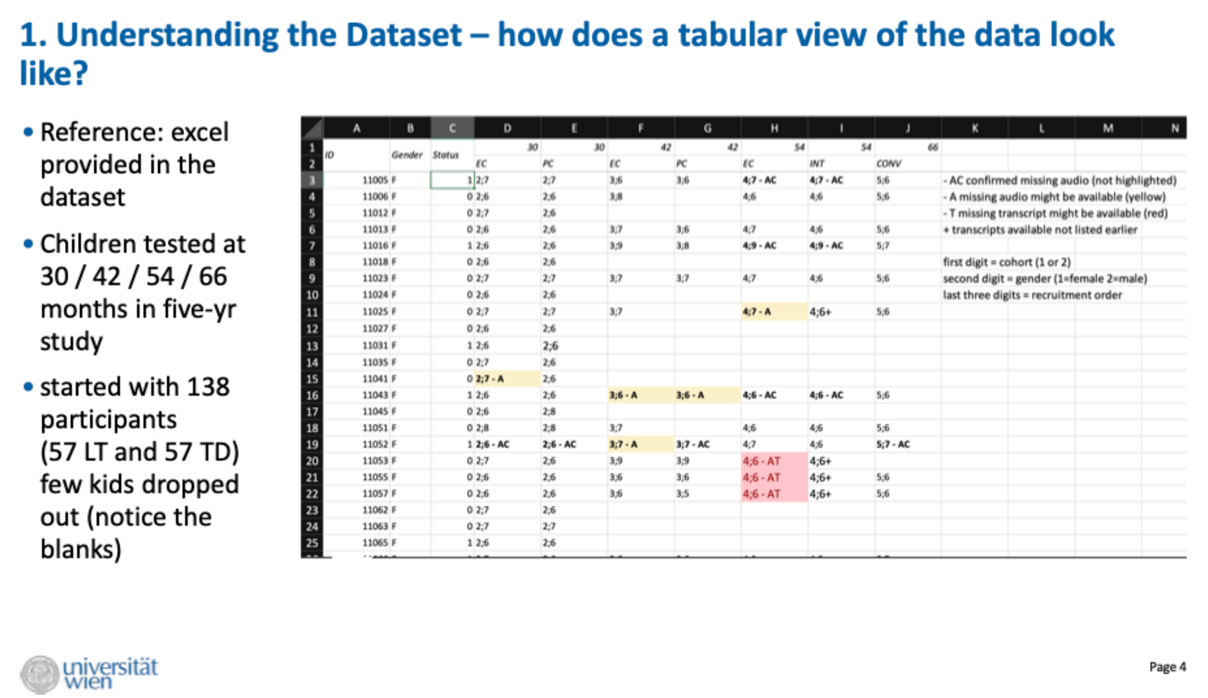
The corpora consists of `.cha` / `.xml` files (depending on which tool you use to extract it) of conversations between an examiner and the child. It is a 5-year longitudinal project, with 56 late talkers and 56 controls with normal language development matched on age, nonverbal cognition, and SES. Some more details about the data (referenced from the corpus, Weismer, 2013):

- Language sample collection was performed at the yearly assessment visits at ages 2;6, 3;6, 4;6, and 5;6
- At ages 2;6 and 3;6 examiner-child (EC) and parent-child (PC) samples were collected using a standard set of toys - Fisher Price Farm set and Doll House plus people and furniture - as the props for play-based conversations
- At age 4;6 an examiner-child (EC) play sample was collected as well as structured interview between the examiner and child (INT)

- Language sampling at 5;6 involved a conversation (CON) between the examiner and child



a visual reference for how the original data looks like



	A	B	C	D	E	F	G	H	I	J	K
1	ID	Gender	Status	30	30	42	42	54	54	66	
2				EC	PC	EC	PC	EC	INT	CONV	
3	11005	F	1	2;7	2;7	3;6	3;6	4;7 - AC	4;7 - AC	5;6	- AC confirmed missing audio (not highlighted)
4	11006	F	0	2;6	2;6	3;8		4;6	4;6	5;6	- A missing audio might be available (yellow)
5	11012	F	0	2;7	2;6						- T missing transcript might be available (red)
6	11013	F	0	2;6	2;6	3;7	3;6	4;7	4;6	5;6	+ transcripts available not listed earlier
7	11016	F	1	2;6	2;6	3;9	3;8	4;9 - AC	4;9 - AC	5;7	
8	11018	F	0	2;6	2;6						first digit = cohort (1 or 2)
9	11023	F	0	2;7	2;7	3;7	3;7	4;7	4;6	5;6	second digit = gender (1=female 2=male)
10	11024	F	0	2;6	2;6						last three digits = recruitment order
11	11025	F	0	2;7	2;7	3;7		4;7 - A	4;6+	5;6	
12	11027	F	0	2;6	2;6						
13	11031	F	1	2;6	2;6						
14	11035	F	0	2;7	2;6						
15	11041	F	0	2;7 - A	2;6						
16	11043	F	1	2;6	2;6	3;6 - A	3;6 - A	4;6 - AC	4;6 - AC	5;6	
17	11045	F	0	2;6	2;8						
18	11051	F	0	2;8	2;8	3;7		4;6	4;6	5;6	
19	11052	F	1	2;6 - AC	2;6 - AC	3;7 - A	3;7 - AC	4;7	4;6	5;7 - AC	
20	11053	F	0	2;7	2;6	3;9	3;9	4;6 - AT	4;6+		
21	11055	F	0	2;6	2;6	3;6	3;6	4;6 - AT	4;6+	5;6	
22	11057	F	0	2;6	2;6	3;6	3;5	4;6 - AT	4;6+	5;6	
23	11062	F	0	2;7	2;6						
24	11063	F	0	2;7	2;7						
25	11065	F	1	2;6	2;6						

excel provided by the researcher that documents each child and their trajectory in the longitudinal study

Methods

Two different `Python` libraries `PyLangAcq and NLTK` were used to process language data. While `NLTK` is the leading tool for working with language data, `PyLangAcq` was a pleasant discovery that happened by chance while working on the project. An important lesson that comes out of this for *any* data science project when working with new forms of data is to analyse all existing open-source libraries available *before* narrowing down on the tool because each library has it's own strengths. While evaluating options, be sure to focus on how well the library is being maintained (using indicators like github installs, recency of commits, and # of stars, etc.) Using a combination of multiple tools could also help in establishing veracity of the results.

An ancillary outcome of the project is that I am able to do a methods-comparison between both these libraries for analysis of language data:

- While `PyLangAcq` is specialised for working with child language acquisition projects, it also supports normal language data analysis very well
- The wrapper objects built in `PyLangAcq` are better than the `NLTK` corollary for the CHILDES corpus in terms of giving one greater control over the chat conversation and being able to intuitively work it as a Python object (this makes NLTK a bit more cumbersome to work with; for example, one has to specify the file path of the particular xml they are interested in, and it's wrapper cannot represent the folder which represents a bunch of xml files, which I had to for each conversation type)
- More number of standard developmental measures can be obtained out-of-the-box using `PyLangAcq` than `NLTK` (such as type-token-ratio, Index of Productive Syntax, etc.)
- The Mean Length of Utterance (Morphemes) values being calculated by `PyLangAcq` are on the higher side. `NLTK` is more reliable for this measure
- There are more functions in `NTLK` to work with raw language data, plus at the utterance and sentence-level

Standard Developmental Measures

The standard developmental measures that have been used to analyse the difference in language production between the two groups have been described below, followed by which of the ones can be reliably used to assess language production as per the current literature in child language acquisition.

1. **Mean Length of Utterance (Morphemes and Words)**
    - Mean length of utterance, or MLU, refers to the average length of the sentences that a child typically uses

- Based on the idea that **each new element of grammatical knowledge contributes to the increase in the length of a child's sentences**
- For example, when children are first learning to talk, their MLU is often 1 because they only use one word at a time: “ball?”, “mommy”, “mine”, “no”. If a child uses a single word like this about half of the time but puts two words together the other half of the time (like “my ball”), then we would say the MLU is 1.5 words
- To calculate MLU, we measure how long a child’s sentences are in either words or morphemes
- Difference between a word and a morpheme:
    - a morpheme is the smallest unit of language that holds its own meaning
    - For ex., jumping: 1 word but 2 morphemes [‘jump + the suffix ‘-ing’] (Speech and Language Kids, 2025)

2. **Type-Token Ratio (TTR)**
    - Type token ratios (TTR) are a measurement of linguistic diversity
    - Defined as the ratio of unique tokens divided by the total number of tokens. This measurement is bounded between 0 and 1
    - TTR scores the lowest value (tendency to use the same words) in conversation
    - Function words are the filler words of a language, such as pronouns, prepositions, and modifying verbs, that fit around the content of a sentence (Deepala, 2025)
3. **Index of Productive Syntax (IPSyn)**
    - The IPSyn, created by Hollis Scarborough to measure syntactic development in preschool-age children. Of available measures of syntax for preschool children, the Index of Productive Syntax is one of the most widely used (Scarborough, 1990)
    - Calculated by **counting the occurrences of 56 syntactic and morphological forms, yielding a total score** and sub-scores for noun phrases, verb phrases, questions/negations, and sentence structures

Results and Conclusion

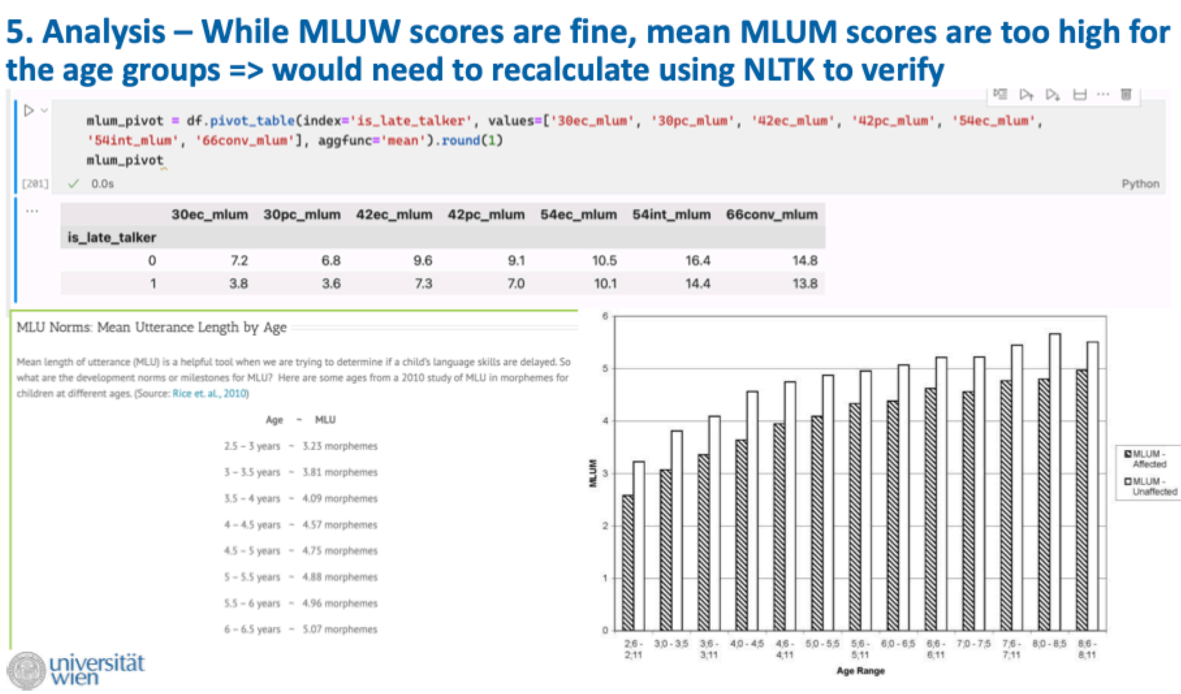
Out of the standard developmental measures discussed above, the type-token-ratio cannot be used as we are comparing texts of different lengths in which children of typical development have usually spoken more utterances. Also, it has been empirically found that TTR scores the lowest value in conversations (Deepala, 2025). Further, while Index of

Productive Syntax is an excellent measure to understand how well a child grasps language at the structural level, it does not help our case. If we use the marker, we would be assessing structural language comprehension and moving away from our research question which focuses on language production.

MLU: A Reliable Marker for Assessment of Language Production

Rice et al., 2010 have demonstrated the reliability and validity of MLU as an index of normative language acquisition and a marker of language impairment through a study which reports age-referenced MLU data from children with specific language impairment and children without language impairments. They used 306 child participants, ages 2;6–9;0 years, 170 with SLI and 136 control children. In the study, 1564 spontaneous language samples were collected, transcribed and analysed for sample size and MLU in words and morphemes. The results show an age progression in MLU words and morphemes, and a persistent lower level of performance for children with SLI (specific language impairment). Thus, we can use MLU as a measure to support our analysis and conclusions.

Further, it was observed that MLU Morpheme scores calculated by `PyLangAcq` were higher in absolute numbers than the typical MLU Morpheme scores found at the age corresponding age groups:



Thus, `NLTK` scores were calculated to verify the scores, in which it was observed that the NLTK provided scores which resemble the typical values more closely.

A main result of the visualisations (bar, box, line and violin plots) and analysis (trend analysis and the difference in absolute scores) is a noticeable difference between the MLU (both morphemes and words) between children of typical development (TD) vs. those who are late talkers (LT), in that the mean scores of TD surpass that of LT by a significant margin in most conversation formats across 2;6 to 5;6 years of age (except for the 4;6 examiner-child format, where they converge; the reason behind this convergence could not be explained). The conclusion supports the existing literature on child language acquisition and shows that MLU (morphemes or words) can be reliably used to assess the language production capacities of a child.

Points for Future Research

- Another tool that could've been helpful for analysing this kind of language data was the `childes-db` that aims to make preprocessing simpler (Sanchez et al., 2019): https://langcog.github.io/childes-db-website/index.html
- A useful analysis of the late talker and typical development statistics could be conducted through building a supervised classifier model which takes as input the developmental measures of both groups of children at 2;6 → and tries to predict the children who would be late talkers at a later age. The accuracy of the model could then be compared by matching it with the classification in the data

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